Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 1 of 8 · Question 7 part b.ii · 3 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

7 (a)	(i)	• volume/concentration of enzyme/lipase; • volume/concentration of indicator; • volume/same type of olive oil; • volume/mass of bile saltes	Ignore amount for all mps	3
	(ii)	40°C;		1
	(iii)	no values given between 30°C and 40°C; no values given between 40°C and 50°C; optimum could be between temperature intervals given/below/above 40ºC/50ºC;		3
(b)	(i)	• tube 3 = 11.5 hours and tube 8 = 24 hours; • 24 - 11.5 = 12.5 hours;	Ecf incorrect values in a subtraction calculation with a correct answer 1 mark	2
	(ii)	• lipase digests/breaks down olive oil; • produces fatty acids; • these acids lower the pH;		3
(c)		• no bile salts so no emulsification of oil/oil not broken down to droplets; • less surface area (for lipase to work on); • lipase only working very slowly/not very active/takes longer to digest oil;		3

Total 15 marks

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 1 of 8 · Question 7 part b.ii · 3 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

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Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 2 of 8 · Question 2 part b.iii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.iii in this mark scheme extract

Number				
2	(a) (i)	W large intestine; X oesophagus; Y stomach; Z small intestine;	Allow colon Allow ileum	4
	(ii)	In the following order only: • bolus; • muscles; • contract; • peristalsis;		4
	(b) (i)	salivary gland/ pancreas;		1
	(ii)	(lining of) **small** intestine/ ileum/ duodenum;		1
	(iii)	• starch digested/broken down to sugar/glucose; • (sugar) diffuses/moves out through (Visking tubing);	Allow maltose	2
	(iv)	Benedict's solution/reagent;		1
	(v)	Any two from the following: • blue/black colour; • (undigested) starch is present; • too large to pass through Visking tubing;		2

Total marks 15

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 2 of 8 · Question 2 part b.iii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 3 of 8 · Question 3 part b.ii · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
3	(a)	(i)	B: A, C and D are not correct as bile is not produced in the gall bladder, small intestine or stomach.		1
		(ii)	Three from: • emulsifies/emulsification fat/lipid; • increases surface area (of fat/lipid); • for lipases/enzymes to digest/work; • neutralises the contents from the stomach;		3
	(b)	(i)	• it contains starch; • it contains protein; • it does not contain reducing sugars/glucose;		3
		(ii)	• contains amylase/carbohydrase; • digested/breaks down the starch; • into glucose/maltose;		3
	(c)		Five from: • peristalsis; • of the food bolus; • (circular) muscles (behind the bolus) contract to push it along; • fibre is indigestible/not broken down; • adds bulk; • stimulates muscles to contract;		5

Total 15 marks

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 3 of 8 · Question 3 part b.ii · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 4 of 8 · Question 2 part b · 3 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number					
2	(a)	(i)	• axes labels; • plots; • scale;	axes correct way round use aty least half of the graph paper for the plotted results ignore extrapolation of line	3
		(ii)	line of best fit;		1
		(iii)	As time increases pH decreases;	ECF from an incorrectly drawn graph	1
	(b)		Any three from: • lipase; • digests/breaks down fat (in milk); • to fatty acids; • which lowers pH/increases acidity;		3
	(c)		time;	allow mins/minutes	1
	(d)		Any two from: • temperature; • volume/concentration of enzyme/substrate (solution)/volume of milk; • type of milk;	ignore amount	2
	(e)		repeat the experiment		1

Total Marks 12

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 4 of 8 · Question 2 part b · 3 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 5 of 8 · Question 4 part b.i · 3 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

4	(a)	(i)	• linear scale/plotted on half of grid; • correct way round; • axes labelled with units; • all points plotted correctly;	mp4- half-square tolerance	4
		(ii)	draw curve of best fit;	Ignore beyond the 2 extremes when line is drawn. 2 straight lines=0 Slight curve at 4 is enough for 1 mark	1
	(b)	(i)	• pepsin digests/breaks down protein; • (to)amino acid/peptides; • (which is)soluble/dissolves; • black silver salts removed /fall away/no longer present;	Mp1- accept hydrolyse MP4- accept silver salts broken down	3
		(ii)	• 1//2/1 to 2; • optimum pH for pepsin;		2
		(iii)	• temperature; • area of film/size/length/width/thickness/; • volume of (pepsin) solution; • type of photographic film/amount of protein in film; • pH;	Mp3-ignore amount	3
	(c)		• adding more enzyme/pepsin has no effect on reaction/ maximum rate of reaction at 4%; • all protein/substrate digested/ substrate is a limiting factor/not enough protein to digest; • enzyme no longer a limiting factor/ active sites empty;	MP1- accept Vmax not enough/no more substrate/protein to bind to active sites= 2 marks	2
					Total = 15

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 5 of 8 · Question 4 part b.i · 3 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 6 of 8 · Question 7 part c · 3 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

7(a)	Any four from: • volume of lactose/sugar solution used; • concentration of lactose/sugar solution; • volume of enzyme solution used; • concentration of enzyme; • type of enzyme used; • how to measure amount/concentration of lactic acid / apparatus to measure amount / concentration of lactic acid;	Allow amount	**(4)**

Question Number	Answer	Additional guidance	Mark
7(b)(i)	25 °C;	accept 20 °C to 45 °C	**(1)**

Question Number	Answer	Additional guidance	Mark
7(b)(ii)	Any two from: • too few results/temperatures investigated / temperatures are too far apart/large gaps between temperatures; • peak/maximum lactic acid production could be below 25°C / could be above 25°C;		**(2)**

Question Number	Answer	Additional guidance	Mark
7(c)	Any three from: • little/no increase in lactic acid concentration / lactic acid concentration would reach a maximum/peak/become constant; • enzyme denatures (due to decrease in pH); • all substrate converted/broken down/used up; • so reaction /slows down/stops;		**(3)**

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 6 of 8 · Question 7 part c · 3 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
7(d)	Any five from: • not possible (to be completely certain); • due to (low) number of pH/temperature values tested/more pH/temperature values need to be tested; • pH5 appears to be optimum pH; • because most sugar broken down/converted at this pH; • optimum pH could be between 5-8/2-5; • optimum temperature not known; • no repeats carried out/no means calculated; • anomalies not identified; • there could be other components in milk that increase enzyme activity/control variables not taken into account;	Allow value stated within ranges given	**(5)**

Total 15 Marks

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 6 of 8 · Question 7 part c · 3 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 7 of 8 · Question 3 part a.iv · 3 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

Number				
3 (a)	(i)	• time in minutes; • 0-10, 11-20, 21-30, 31-40,41-50,51-60;		2
	(ii)	• 75 + 72 + 68; plus/minus 2 for each one • 215 cm^3;	ECF	2
	(iii)	• rapid reaction/a lot/large amount of gas produced, at start; • rate slows/less gas (for next 10 minutes); • drops a lot/little gas produced (after 40 minutes);		3
	(iv)	any three from • at start plenty of substrate/many E-S complexes formed; • substrate used up/less substrate available; • reaction slows/less gas/product, produced/less enzyme action; • most substrate used at end so little gas produced;		3
(b)		• less gas produced/reaction lower/less; • because active sites blocked;		2

Total 12

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 7 of 8 · Question 3 part a.iv · 3 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 8 of 8 · Question 4 part c · 2 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

4	(a)	(i)	• linear scale/plotted on half of grid; • correct way round; • axes labelled with units; • all points plotted correctly;	mp4- half-square tolerance	4
		(ii)	draw curve of best fit;	Ignore beyond the 2 extremes when line is drawn. 2 straight lines=0 Slight curve at 4 is enough for 1 mark	1
	(b)	(i)	• pepsin digests/breaks down protein; • (to)amino acid/peptides; • (which is)soluble/dissolves; • black silver salts removed /fall away/no longer present;	Mp1- accept hydrolyse MP4- accept silver salts broken down	3
		(ii)	• 1//2/1 to 2; • optimum pH for pepsin;		2
		(iii)	• temperature; • area of film/size/length/width/thickness/; • volume of (pepsin) solution; • type of photographic film/amount of protein in film; • pH;	 Mp3-ignore amount	3
	(c)		• adding more enzyme/pepsin has no effect on reaction/ maximum rate of reaction at 4%;	MP1- accept Vmax	2
			• all protein/substrate digested/ substrate is a limiting factor/not enough protein to digest; • enzyme no longer a limiting factor/ active sites empty;	not enough/no more substrate/protein to bind to active sites= 2 marks	
					Total = 15

Type MS · Explain an enzyme practical result, including levelling off as substrate runs out

Answer 8 of 8 · Question 4 part c · 2 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks